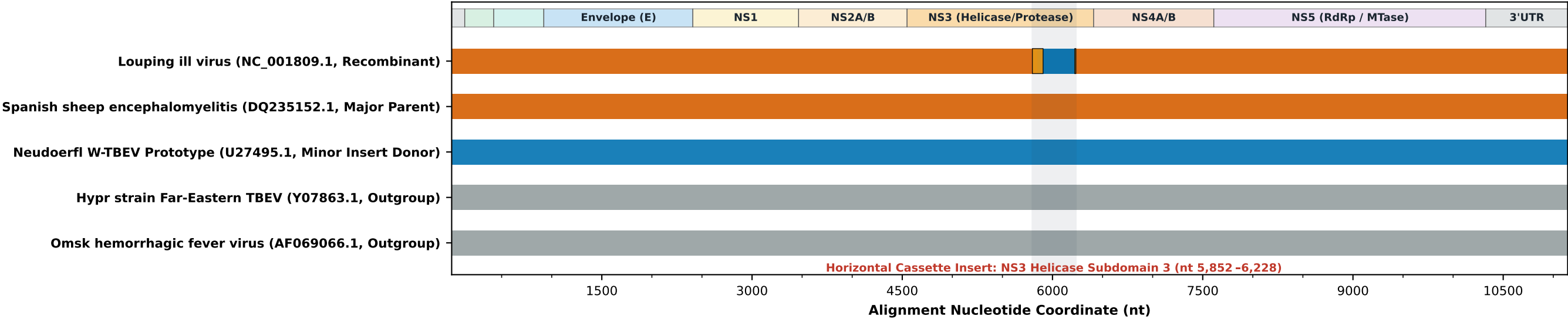
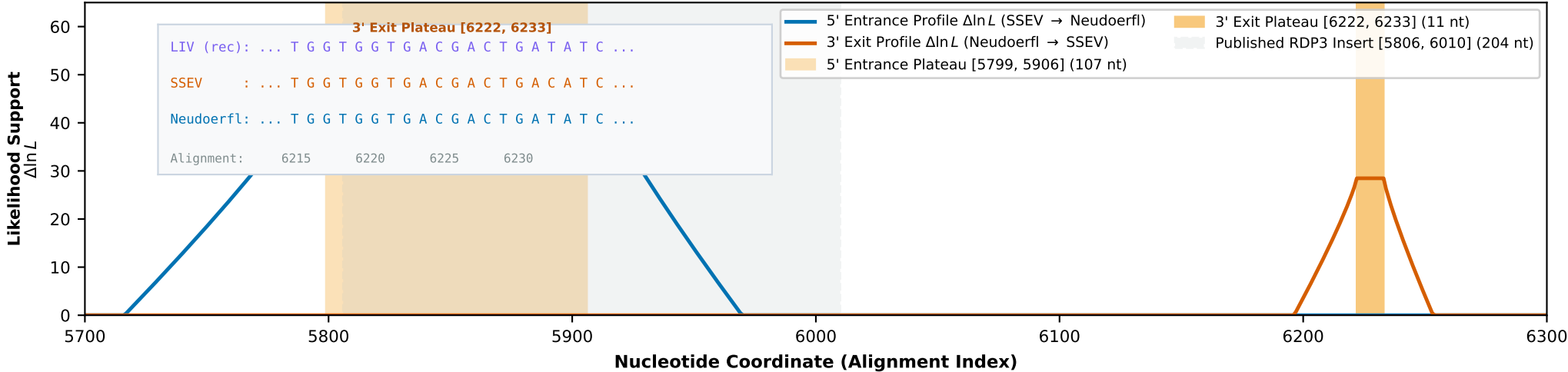


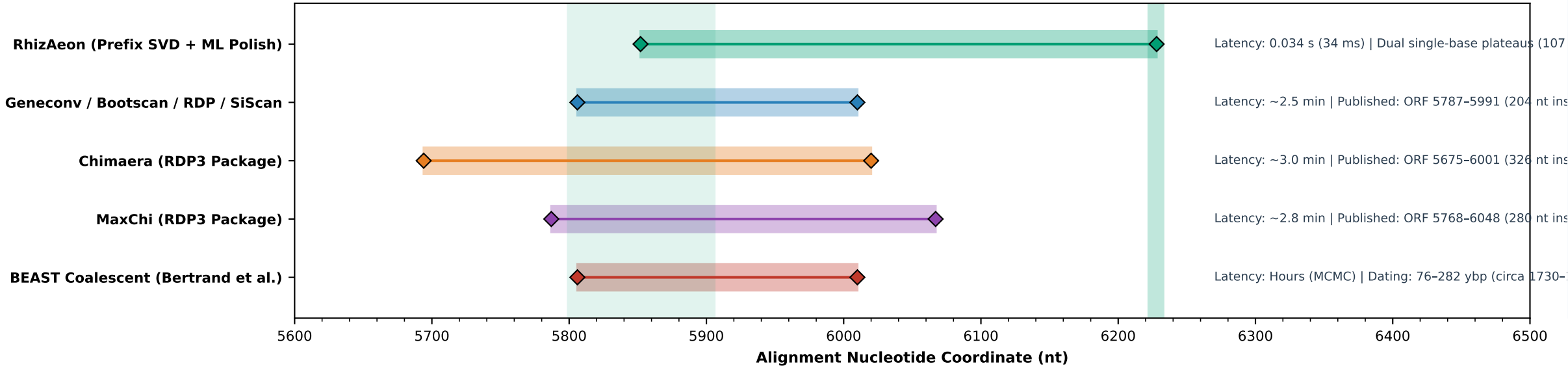
Track 1: Cohort Mosaic Horizon (Tick-Borne Flaviviruses: Louping Ill Virus Recombinant)



Track 2: Physical Mapping Uncertainty & Dual Likelihood Plateaus Across NS3 Helicase Cassette



Track 3: Multi-Tool Concordance & Disagreement Horizon (#, Location, Runtime)



STUDY DOSSIER

Flavivirus Louping Ill

Bertrand et al. (2012) PLOS ONE

DOI: [10.1371/journal.pone.0031981](https://doi.org/10.1371/journal.pone.0031981)

ALIGNMENT PROVENANCE

• File: tbev_loupingill_aligned.fasta

• Taxa Analyzed: 6 sequences

• Alignment Length: 11,144 nt

• Segregating SNPs: 2,631 sites

• SHA-256 Hash Verified:

72899ddb6a63f0868047...

STATUS & CORROBORATION

✓ CONFIRM & REFINE

• Recombinant: LIV (NC_001809.1)

• Major Parent: SSEV (DQ235152.1)

• Minor Parent: Neudoerfl (U27495.1)

• **5' Entrance: nt 5,852**

Plateau: [5799, 5906] (107 nt)

Contains Pub. RDP nt 5,806

• **3' Exit: nt 6,228**

Plateau: [6222, 6233] (11 nt)

Flank: Neud:6222 | SSEV:6234

• Support $\Delta \ln L$: +51.2 / +28.5

COMPUTATIONAL SCALING

• **RhizAeon Latency: 34.3 ms (0.034 s)**

• RDP3 Package: ~2.5-3.0 min

• BEAST Coalescent: Hours

• **Acceleration: >1,000x vs RDP3**

BIOLOGICAL NARRATIVE

• Confirms authentic horizontal cassette

insertion in NS3 helicase of LIV.

• Resolves 112-nt RDP3 heuristic discrepancy

(Chimaera 5675 vs Geneconv 5787):

wide 107-nt plateau unifies 5' entry,

while 11-nt plateau pins 3' exit.